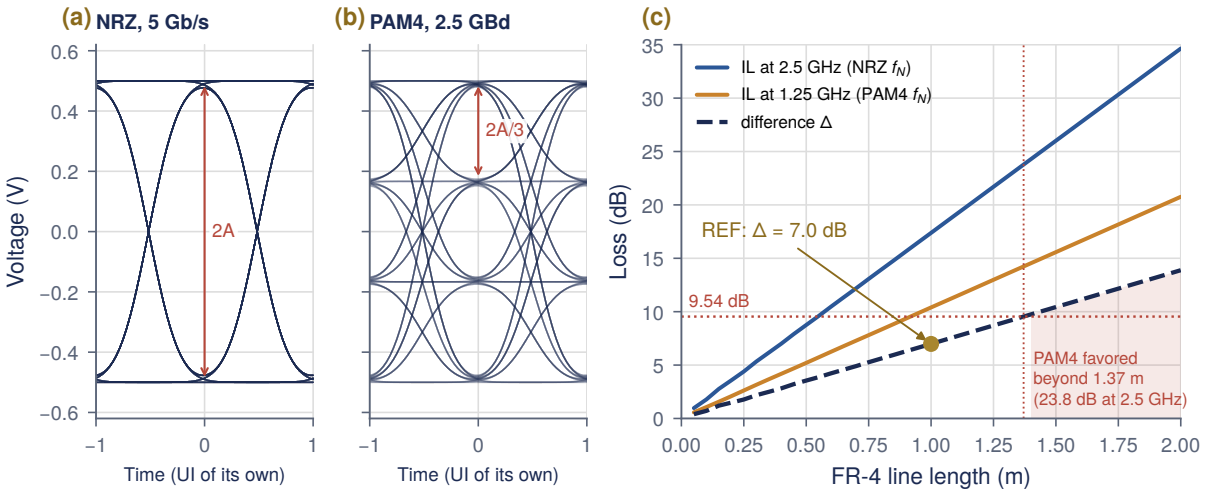

(a) NRZ, 5 Gb/s
(b) PAM4, 2.5 GBd
(c)
Voltage (V)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
2A
2A/3
−1
0
1
Time (UI of its own)
Time (UI of its own)
IL at 2.5 GHz (NRZ f_N)
IL at 1.25 GHz (PAM4 f_N)
difference Δ
REF: Δ = 7.0 dB
9.54 dB
PAM4 favored
beyond 1.37 m
(23.8 dB at 2.5 GHz)
Loss (dB)
35
30
25
20
15
10
5
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
FR-4 line length (m)